SIMATS ENGINEERING

ASSESSMENT TOOL **1**

COURSE NAME: Fundamentals of Data Science

COURSE **CODE**: **DSA04**

COURSE OUTCOME COVERED

CO1: Analyze the role of data science, facets of data, and the big data ecosystem for informed *decision*-making. ***(BL2)***

QUESTIONS: 20

Annexure A

SHORT ANSWER TEST

Total Marks:100

***Code* of *Conduct*:**

1 M Ayesha / **1923** 241207/Name / ***Reg*** No) ***certify that* this *submission* is my original** work *and that I have adhered to the* guidelines ***specified*** for this *assessment*.

I understand that any violation of academic **integrity** rules will result in disciplinary action.

Signature of the student**:**

Annexure A

Short Answer Test

1. A college wants to identify students who may fail in the final exam. How can Data Science help?

2. A hospital has patient records for many years. How can Data Science be applied? 3. An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?

4. A bank wants to detect suspicious transactions. How can Data Science be useful? 5. A company wants to reduce customer loss. How can Data Science support this?

6. A school wants to improve student performance. Mention one benefit of Data Science in this case.

7. A supermarket wants to know which products sell more during festivals. How can Data Science help?

8. A transport department wants to reduce traffic congestion. How can Data *Science* be applied?

9. A farmer wants to improve crop yield. Give one use of Data Science.

10. A company has data in Excel tables, customer reviews, images, and videos. Which facet of data is shown here?

11. A weather app updates temperature every minute. Which facet of data is involved? 12. A university stores student details in rows and columns. What type of data structure *is* this?

13. Customer comments on social media are collected for analysis. What type of data is this?

14. A company collects a very large amount of daily user activity data from its mobile

app. Which Big Data characteristic is applied?

15. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?

16. A business wants to convert customer data into useful sales decisions. Which Big Data characteristic is important?

17. A company stores large files in Hadoop Distributed File System. Which part of the Big Data ecosystem is used?

18. A company wants to process large datasets using parallel computation. Which Big *Data* tool can be used?

19. An organization wants to collect live sensor data from IoT devices. Which Big Data ecosystem component is suitable?

20. A company wants to visualize Big Data results for decision-makers. Which tool or method can be applied?

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4- Exemplary (5 Marks)	Level 3- Proficient (4 Marks)	Good
Conceptual Understanding	30	Demonstrates clear and complete		

Accuracy of
Answer

25

Application of
Knowledge

Completeness of
Response



with minor **gaps**

15

Clarity **&**
Presentation

10

understanding of **the**
understanding
concept

Completely correct Mostly
correct answer with no **errors with** minor
errors

Correctly
applies
concepts to solve **the with minor** question/problem

Covers all required
parts **of the question**

Clear**,** concise, and
well-organized
answer

with minor
omissions

Partially correct **with** noticeable errors

Mark)

Poor or incorrect understanding

Incorrect or **rrelevant** answer

Understandable Somewhat unclear or Difficult **to with minor** issues poorly **structured**

understand

Level 2- Developing (2-3 Marks)

Level 1- Beginning (0-1

Basic understanding with some misconceptions

Applies concepts

Limited or **partially** mistakes

correct application

No or incorrect application

Covers most parts

Covers **only** some

Incomplete or **parts**

missing response

1.A hospital has patient records for many years.

How Can Data Science be applied?

Hospital Patient Records

* Data Science Can analyze historical Patient data to identify disease Patterns and Predict health risks. * It can also help doctors make batter treatment decisions.

2 A bank wants to detect suspicious transactions. How Can Data Science be useful?

Bank - Suspicious Transactions

* Data Science can analyze transaction Patterns to identify unusual or Suspicious activities.

* It helps banks detect and Prevent fraud.

3. A School - wants to improve Student Performance.

Mention

one benefit of Data Science in this case.

School - Student

Performance

-

* Data Science can analyze Student's marks, attendance

and leavning Patterns.

* It helps identify week Students and. Provide them with

Proper Support.

4. A transport department wants to reduce traffic.

Congestion How can Data Science be applied?

Transport - Traffic Congestion

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* Data Science Can analyze traffic data, vehicle movement, and road Conditions.

* It helps Predict traffic congestion and improve routes and traffic Signals":

5. A Company has data in Excel tables, Customer *reviews,* images, and videos, which facet of data is shown here?

Different Types of Data

* This shows the variety characteristic of Big Data. * Data is available in defferent forms such as Excel, tables, text, images, and videos.

6. A university Stoves Student details in rows

and Columns. what type of data Structure is this?
University - Student Details aren

* This is Structured data because Student information

is organized in rows and columns.

* It can be easily stored and queried in a database 7 A Company Collects a very large amount of daily user activity data from its mobile app. which Big Data characteristic is applied?

Large Amount of Data

* This represents the volume characteristic of

Big Data.

* It refers to the huge amount of data generated

and Collected every day.

8. Useful Business Decisions:-

* The important Big Data Characteristic is value. * It means extracting usefull information from

data to Support business decisions.

9. Parallel Processing :-

* Apache Spark can be used to process large datasets

Using Parallel Computation

* It divides the work across multiple computers or

Processors.

10. Big Data

Visualization

* Data Visualization tools such as Tableau or Power BI Can be used to present Big Data results.

* They help decision-makers understand Patterns and make better decisions.